

Item Specifications and Instructions for CUP MIOX Vault AMB60SC Replacement Cell

1.0 Background

The Central Utility Plant (CUP) or Steam and Chilled Water Generation Plant (S&CWGP) is a 24/7 operation that delivers steam, chilled water and compressed air to the National Institute of Standards and Technology (NIST) at Gaithersburg Campus. To prevent damage to the CUP equipment, piping and site systems; it is imperative that NIST maintain a proactive treatment program.

The CUP facility uses a MIOX on-site chemical generator to treat the recirculated cooling tower (condenser) water for the Chiller units. The MIOX equipment uses salt and electricity to generate on-demand disinfection chemicals, such as algaecides and biocides, to treat and control the spread of harmful microorganisms, bacteria (such as Legionella), viruses, fungi and algae in the recirculated cooling tower (condenser) water.

Presently, the NIST CUP MIOX on-site chemical generator is out of service, and needs a new replacement cell.

2.0 Objectives

The objective is to provide one (1) new Vault 60 cell for the existing NIST Central Utility Plant MIOX on-site chemical generator unit.

The cell component shall be new, and shall **not** be a prototype, a demonstration unit, or a used or refurbished unit. The supplier shall specify the warranty period of the new unit prior to purchase.

The cell component will be installed by NIST personnel. No installation services are required under these specifications and instructions.

3.0 Specifications

The specifications for the replacement 60 cell unit must match (or exceed) and be compatible with the existing MIOX 60 cell component being replaced. Reference Attachment 1 and 2.

3.1 Item Specifications

The existing MIOX Vault AMB60SC cell information:

- Description: Cell assembly M60SC
- DNWT Legacy Part Number: 201-01304-01 (original part number)
- DNWT SAP Number: 47000137 (new part number)

The MIOX Vault AMB60SC replacement cell shall meet or exceed the following specifications:

- Rated Free Available Chlorine (FAC): 60 lb./day
- 12.5% Bleach Equivalent: 60 gal/day
- Water Treatment Capacity: 7.2 mgd
- Salt Conversion (SCE): 3.0 lb. salt /lb. FAC
- Energy Conversion (ECE): 3.0 kW-hr/lb. FAC
- FAC Concentration: 4,500 mg/L (+/- 1,000 mg/L)

Specifications and Instructions for CUP MIOX Maintenance and Repair

- Flow Rate (+/- 15%): 76 gal/hr.

3.2 Unit Information:

The existing on-site water disinfection unit that requires the replacement cell is the following (reference Attachment 2):

- Brand: MIOX (De Nora)
- Part Number: 000-00171-05
- Serial Number: 1414VM60ABS0017
- Model: Vault ABM60SC
- Voltage: 200-240 V
- Frequency: 50-60 Hz
- Current: 120 A
- Interrupt: 10 KA
- Conductors: Only copper conductors, 1/0 AWG – 53.46MM² – 120 Amps

4.0 Delivery Schedule

The lead time for delivery shall not exceed six (6) weeks after receipt of order/award.

5.0 Place of Delivery

National Institute of Standards and Technology (NIST)
100 Bureau Drive
Building 301 Shipping and Receiving
Gaithersburg, Maryland, 20899.

6.0 Deliverable

The cell component shall be delivered FOB Destination.

FOB Destination means: The Contractor shall pack and mark the shipment in conformance with carrier requirements, deliver the shipment in good order and condition to the point of delivery specified in the purchase order, be responsible for any loss of and/or damage to the goods occurring before receipt and acceptance of the shipment by the consignee at the delivery point specified in the purchase order; and pay all charges to the specified point of delivery.

7.0 Inspection and Acceptance Criteria

The Government (NIST) shall inspect the Cell component upon arrival and notify the supplier if there are any discrepancies in the shipment. Final acceptance of the equipment shall be done once received and inspected for damage and complete part in accordance with FAR Clause 52.212-4(a) *Inspection/Acceptance*.

8.0 Security

Vendor or private courier delivering to NIST shall access the site through Gate “F” on Muddy Branch Road for deliveries and Truck inspection. The contractor must provide the name(s) of the courier at least one (1) week in advance of delivery to the designated NIST point of contact(s), so that the courier can be register in the NIST Visitor system.

9.0 Invoicing

Invoices shall be submitted for payment upon **acceptance** of the replacement cell for the MIOX unit. Invoices shall be accurate, up-to-date, free from duplications, and include the information listed under the NIST Clause for Electronic Billing Instructions. Invoices shall be sent to accounts payable at NIST as directed in the NIST Clause.

Specifications and Instructions for CUP MIOX Maintenance and Repair

10.0 Warranty

The supplier shall provide and abide by the manufacturer's warranty as agreed to in the contract.

11.0 Attachments

Attachment 1 – NIST MIOX Front Cabinet Photo

Attachment 2 – NIST MIOX Name Plate Information Photo